

Qijun Wu

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EDUCATION

Shenzhen University <i>Incoming M.S. Student in Applied Statistics (admitted through recommendation)</i>	Expected enrollment: Sep. 2027
Guangzhou University <i>B.S. in Statistics</i>	Sep. 2023 - Jun. 2027 (Expected)
• Academic performance: GPA 3.78/4.00; average score 90.73; ranked 7th of 72 students (top 10%); CET-6. • Relevant coursework: Probability Theory (91), Mathematical Statistics (91), Statistical Computing (96), Multivariate Statistics (93), Data Mining (92), Regression Analysis (98), Time Series Analysis (91), Data Structures & Algorithm Analysis (99), Mathematical Analysis Topics (98), and Advanced Algebra Topics (99).	

RESEARCH INTERESTS

My research interests include **AI for Science (AI4S)**, **AI for Statistics**, and **high-dimensional statistical inference**. I am particularly interested in AI-assisted scientific computing, GPU-based statistical simulations, and their validation. My statistical work focuses on **high-dimensional inference** and **interval estimation**, especially Type I error control, test power, coverage, and interval length, studied through theory and simulation.

RESEARCH EXPERIENCE

Power-Enhanced Tests for the High-Dimensional Two-Sample Behrens-Fisher Problem	Sep. 2025 – Present
• Developed WCCT by integrating Welch’s t-tests with the Cauchy combination method, then proposed EWCT and EWCTs as power-enhanced variants for sparse, dense, and scale-heterogeneous signals. • Designed simulation studies and real-data analyses to compare the proposed methods with existing approaches in terms of Type I error control and empirical power at small-to-moderate sample sizes. • Led method development, theoretical derivations, experimental design, results analysis, and initial manuscript drafting.	
Prior-Free Probabilistic Interval Estimation for Constrained Parameters	Jul. 2025 – Jun. 2026
• Developed prior-free IM and NIM interval methods for constrained normal and Poisson means with nuisance parameters. • For normal means, IM achieved exact coverage, with intervals slightly longer than Bayesian intervals. For Poisson means, IM was conservative, while NIM achieved coverage closest to the nominal level. • Implemented the methods, analyzed neutrino-mass and signal-intensity data, and drafted the initial manuscript.	
Exact Confidence Intervals for Poisson Signals with Unknown Background	Sep. 2025 – Present
• Developed an exact confidence-interval method for Poisson signals with unknown backgrounds using the h-function, Poisson-quantile truncation, and Berger-Boos correction to address infinite sample spaces and nuisance-parameter optimization. • Compared interval coverage and length through simulations and analyzed KARMEN neutrino-detection data. • Drafted the manuscript, clarified the methodology, and conducted additional experiments.	

PUBLICATIONS

1. Lu, H., & **Wu, Q.** (2026). Power-Enhanced Cauchy Combination Tests for the Two-Sample High-Dimensional Behrens-Fisher Problem. (Under review at *Journal of Multivariate Analysis*)
2. Lu, H., Zhao, S., & **Wu, Q.** (2026). Optimal exact confidence intervals for Poisson signal with unknown background: A quantile-truncated h -function approach with Berger–Boos correction. (Major revision at *Statistical Papers*)
3. Lu, H., & **Wu, Q.** (2026). Prior-free probabilistic interval estimation for constrained parameters. *AIMS Mathematics*, 11(6), 17320–17353. DOI: 10.3934/math.2026709.

PROJECTS

- SimForge-GPU: AI-Assisted GPU Migration and Validation for Statistical Simulations

GitHub
- Developed a local GPU migration tool for **AI coding assistants** and statistical simulation programs. It analyzes scripts, creates conversion plans, and conservatively rewrites supported CPU code to run on GPUs.
 - Used **Python AST analysis** to identify supported function calls and unsupported behavior; reviewed structured suggestions from external AI coding agents, applied constrained API rewrites, and generated traceable artifacts.
 - Implemented repeated **CPU/GPU** output comparisons and runtime measurements, producing JSON/Markdown validation, benchmark, and explanation reports. Added examples and regression tests for **Monte Carlo estimation**.
 - Applied the toolkit to the **IM/NIM Monte Carlo studies** reported in *AIMS Mathematics*. With $M = N = 10,000$, $m = 20$, $\lambda = 1$, $\varepsilon = 3$, and nominal 90% coverage, a Python/PyTorch implementation using batched tensors and vectorized bisection estimated one IM interval in **15 seconds**, compared with **11 hours** using R’s serial **uniroot** implementation.
- Service Supply-Demand Gap Identification for Older Adults in Guangzhou under High-Temperature Conditions

 - Constructed indicators of **heat exposure**, **older-adult vulnerability**, and **service supply**, framing service-gap identification as a **multi-criteria decision-making** problem.
 - Combined **entropy weighting**, principal component analysis (**PCA**), **entropy-weighted TOPSIS**, and **Borda ranking** to assess regional supply-demand differences.
 - Identified **Yuexiu**, **Haizhu**, and **Liwan** as areas with pronounced supply-demand mismatches and developed **policy recommendations** accordingly.

TECHNICAL SKILLS

Programming & Computing: Python, R; GPU-based Monte Carlo simulation.
Statistical Methods: High-dimensional hypothesis testing, interval estimation.
Research Tools: Git/GitHub, LaTeX, Word; literature review and scientific writing.

HONORS & AWARDS

12th National Undergraduate Statistical Modeling Competition (Second Prize , Guangdong Province)	Jun. 2026
The First Prize Scholarship	Sep. 2026
The Third Prize Scholarship	Jan. 2026
15th MathorCup Mathematical Modeling Competition (Third Prize , Guangdong Province)	Apr. 2025
15th National Undergraduate Mathematics Competition (Third Prize , Guangdong Province)	Dec. 2023